

The Allocation of Teaching Talent and Human Capital Accumulation

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Introduction

- ▶ Public education in U.S. has gone through major (positive) changes since end of WW II:
 - Annual real expenditures per student: \$2,100 (1950s) to \$12,000 (2010s)
 - Student-teacher ratio: 27 (1955) to 16 (2010s)
- ▶ Moderate to no changes in levels of educational achievement (NAEP)
- ▶ Potential explanations include:
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 - Role of teachers' unions (seniority vs. merit)
 - **Occupational choice** (other occupations are more attractive)
 - Local funding for public education (e.g. property taxes) amplifies educational disparities

Research Questions

- ▶ To what extent do changes in career opportunities in other occupations affect selection of workers into teaching careers?
- ▶ To what extent are static efficiency gains associated with improved career opportunities in non-teaching occupations muted or amplified by dynamic human capital investment effects?

Outline

Stylized Facts

Model

Data

Calibration Strategy

Benchmark Results

Counterfactual Experiments

Conclusion

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Stylized Fact #1

Majority of (Public) School Teachers is Female

Time Period		% Female
early 70s	Project TALENT	61.1
	Census 1980	59.8
1986-1993	NLSY79	77.7
	Census 1990	74.8
2009-2013	NLSY97	77.1
	ACS 2009-2013	76.4
2003-2004	NCES (2006)	75

Stylized Fact #2

Educational Barriers / Labor Market Discrimination

- ▶ Women face low barriers / discrimination in teaching
- ▶ Barriers / discrimination in non-teaching occupations for women generally falling over time

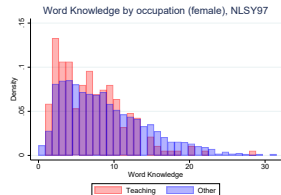
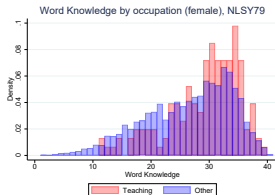
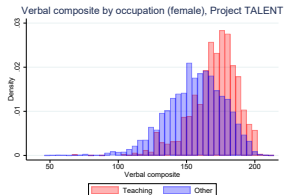
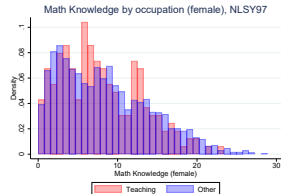
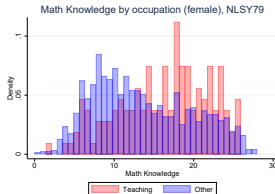
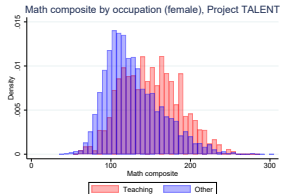
Stylized Fact #3

Trends in Occupational Choice

- ▶ Share of teachers in labor force rising between 1970 and 2010
 - Share of teachers among women: 3.4% in 1970, 6.0% in 2010
 - Share of teachers among men: 2.6% in 1970, 1.9% in 2010
- ▶ Sharp rise in female labor force participation rate between 1970 and 1990
- ▶ Slight decline in male labor force participation rate between 1990 and 2010

Stylized Fact #4

Trends in Ability Distribution of Women by Occupation



Stylized Fact #5

Trends in Education Expenditures and Scholastic Achievement: International Comparison

- ▶ Positive relationship between growth in spending per student and improvements in academic performance on next test across eight high-income nations 1995-2015.¹
Weak relationship for the United States (correlation 0.05 vs. 0.28 for other nations).
- ▶ Weak relationship between changes in pupil:teacher ratio and improvements in academic performance on next test.
Similar relationship in US and abroad (correlation 0.06 vs. 0.03 for other nations).

¹Czech Republic, Italy, Mexico, New Zealand, Portugal, Spain, Sweden & the United States. 

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Model

Three Major Building Blocks

1. Two-period OLG model with human capital investment
2. Agents' occupational choices based on comparative and absolute advantage
3. Educational barriers / labor market discrimination (as in Hsieh et al., 2019)

Model

Endowments, Preferences

- ▶ Each period, a measure $\frac{M}{2}$ of agents is born and lives for two periods: “young” and “old”
- ▶ G groups of individuals indexed by $g \in \{1, \dots, G\}$
- ▶ I occupations indexed by $i \in \{1, \dots, I\}$
- ▶ Occupational abilities $\vec{a}$ drawn from joint distribution $F(\vec{a})$
- ▶ Agents have log preferences over consumption (tomorrow) and leisure (today):

$$\mu \ln c'_g + \ln(1 - s_{i,g})$$

Simple Two-Sector Model

Technologies

- ▶ “Young” make occupation-specific investments in terms of time (s) and goods (e)
- ▶ “Old” work (as **teachers** or **production workers**) and consume

Human capital production (teaching) depends on teacher's human capital $h_{T,\hat{g}}$, class size $N(h_{T,\hat{g}})$, own ability a_i , time $s_{i,g}$ and goods $e_{i,g}$ investments:

$$h'_{i,g}(a_i) = (h_{T,\hat{g}})^\beta a_i^\alpha (s_{i,g})^\phi (e_{i,g})^\eta (N(h_{T,\hat{g}}))^{-\sigma}$$

Final output production depends on adult worker's human capital $h_{O,g}$ and exogenous (occupation-specific) productivity A_O :

$$y_g = A_O h_{O,g}$$

Simple Two-Sector Model

Values

$$V_g(a_T, a_O, \tilde{H}_T) = \max_{\{s_{O,g}, s_{T,g}, e_{O,g}, e_{T,g}\}} \left\{ V_{O,g}(a_O, \tilde{H}_T), V_{T,g}(a_T, \tilde{H}_T) \right\}$$

where

$$\begin{aligned} V_{O,g}(a_O, \tilde{H}_T) &= \ln \left(1 - s_{O,g} \left(a_O, \tilde{H}_T \right) \right) \\ &\quad + \mu \ln \left[h'_{O,g} A'_O (1 - t') (1 - \tau_{O,g}^{\omega'}) - e_{O,g}(a_O, \tilde{H}_T) (1 + \tau_{O,g}^e) \right], \\ V_{T,g}(a_T, \tilde{H}_T) &= \ln \left(1 - s_{T,g} \left(a_T, \tilde{H}_T \right) \right) \\ &\quad + \mu \ln \left[\omega'_{T,g}(h'_{T,g}) (1 - t') (1 - \tau_{T,g}^{\omega'}) - e_{T,g}(a_T, \tilde{H}_T) (1 + \tau_{T,g}^e) \right] \end{aligned}$$

Simple Two-Sector Model

Teachers' Compensation Funded by Labor Income Tax t

$$\begin{aligned} t \left[\sum_{g=1}^G \int_0^{\infty} (1 - \tau_{T,g}^{\omega}) \omega_{T,g}(h_{T,g}(a)) f_{T,g}(a) da + \sum_{g=1}^G \int_0^{\infty} (1 - \tau_{O,g}^{\omega}) A_O h_{O,g}(a) f_{O,g}(a) da \right] \\ = \sum_{g=1}^G \int_0^{\infty} (1 - \tau_{T,g}^{\omega}) \omega_{T,g}(h_{T,g}(a)) f_{T,g}(a) da \end{aligned}$$

where

$$f_{T,g}(a) = \int_0^{\bar{a}_g^{-1}(a)} f(a, b) db \quad \text{and} \quad f_{O,g}(b) = \int_0^{\bar{a}_g(b)} f(a, b) da$$

Simple Two-Sector Model

Law of Motion for $\tilde{H}_T$

$$\tilde{H}'_T = \sum_{g=1}^G \int_0^\infty \left(\left(\frac{2\tilde{H}_T}{M} \right)^\sigma a^\alpha s_{T,g} \left(a, \tilde{H}_T \right)^\phi e_{T,g}(a, \tilde{H}_T)^\eta \right)^{\frac{\beta}{\sigma}} f_{T,g}(a) da$$

Simple Two-Sector Model

Occupational Threshold

For each $g \in \{1, \dots, G\}$ and a_O , there is a teaching ability $a_{T,g}^*(a_O)$ such that

$$V_{O,g}(a_O, \tilde{H}_T) = V_{T,g}(a_{T,g}^*(a_O), \tilde{H}_T)$$

Simple Two-Sector Model

Assignment of Students to Teachers

- ▶ Class size is function of teacher's human capital $h_{T,g}$:

$$N(h_{T,g}) = h_{T,g}^{\frac{\beta}{\sigma}} \cdot \frac{M}{2\tilde{H}_T}$$

- ▶ Class size function implies that:
 - students are indifferent between teachers with different $h_{T,g}$
 - assignment of students to teachers is random
- ▶ Teacher's wage $\omega_{T,g}$ is exogenous function of human capital:

$$\omega_T(h_{T,g}) = \kappa h_{T,g}^{\gamma}$$

Equilibrium

Given occupational choices of today's "old" and aggregate human capital in teaching $\tilde{H}_T$, an equilibrium consists of individual choices of "young" $\{e_{T,g}, s_{T,g}, e_{O,g}, s_{O,g}\}$, the occupational choice boundary $a_{T,g}^*(a_O)$, the corresponding densities $f_{T,g}$ and $f_{O,g}$, and occupation- and group-specific wage profiles $\{\omega_{T,g}, \omega_{O,g}\}$ such that:

1. Individuals solve their investment problems

► Time Investment (s)

► Goods Investment (e)

2. Government budget constraint is satisfied

► Aggregate Law of Motion

► Transition Dynamics

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Data

- ▶ Micro-data on abilities and occupational choice:
 1. Project TALENT (1960-1975):
 - representative 5% sample of high school population in 1960
 - follow-up surveys at 1, 5, and 11-year post graduation
 2. NLSY 79
 3. NLSY 97
- ▶ Respondents' *math*, *verbal*, and *social* abilities based on aptitude tests in all three longitudinal surveys
- ▶ Occupational choice 11 years after (likely) high school graduation in all surveys (~ age 29)

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Calibration

Assumptions, Normalizations, Free Parameters

Parameter	Definition	Determination	Value
$\tau_{o,men}^w$	labor market barriers for men	assumption	0
$\tau_{o,g}^e$	human capital barriers for all groups	assumption	0
$\tau_{T,g}^w = \tau_{T,g}^e$	labor market & human capital barriers in teaching (all groups)	assumption	0
α	elasticity of human capital with respect to idiosyncratic ability	normalization	1
$A_{21,1970}$	productivity in “Home Production” (1970)	normalization	1
β	elasticity of student’s human capital with respect to teacher’s human capital	free parameter	0.5
σ	elasticity of student’s human capital with respect to class size	free parameter	0.5

Calibration

Time-Invariant Parameters

Parameter	Definition	Determination	Value
θ	shape parameter of Fréchet-distributed idiosyncratic abilities	wage dispersion in non-teaching occupations (indirect inference)	1.476
η	goods elasticity of human capital	aggregate education spending share (indirect inference)	0.103
ϕ	time elasticity of human capital	Mincerian returns to education (non-teaching, indirect inference)	2.745
γ	curvature of wage function in teaching	wage dispersion in teaching (indirect inference)	0.83
μ	trade-off between consumption and time spent accumulating human capital	schooling of teachers relative to schooling of others	0.714

Calibration

Time-Varying Parameters

Parameter	Definition	Determination
A_o	occupational productivities (non-teaching)	labor market shares for men
$\tau_{o,women}^w$	labor market barriers (non-teaching) faced by women	labor market shares for women
κ	scale parameter of wage function in teaching	fraction of males who are teachers
λ_f	composite labor market barrier for women in non-teaching occupations	fraction of females who are teachers

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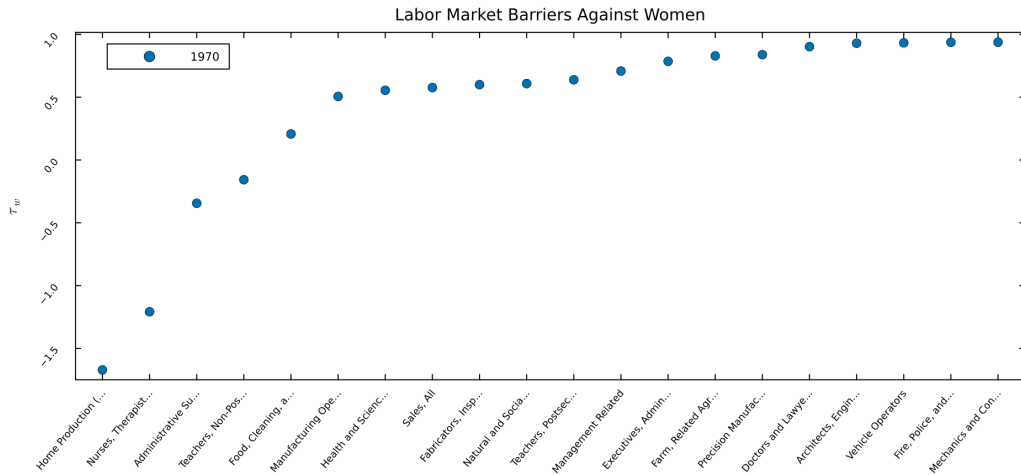
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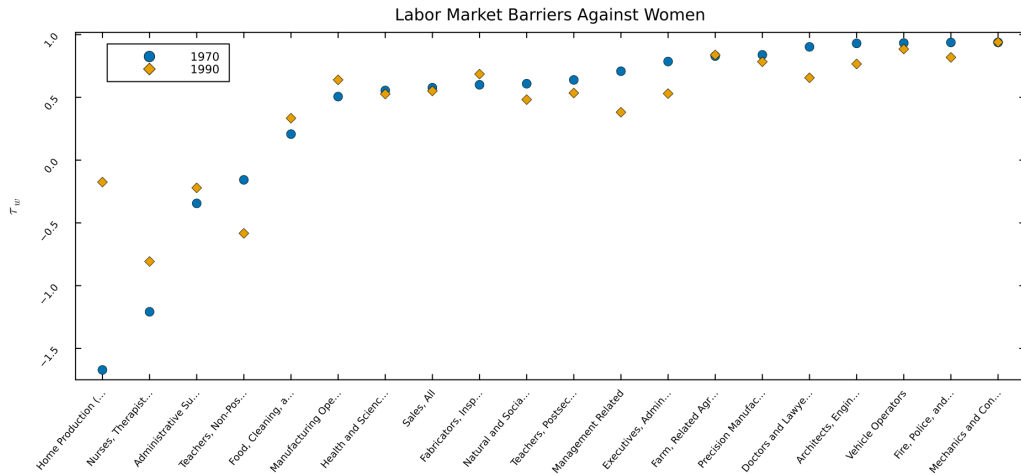
Labor Market Barriers

► List of Occupations



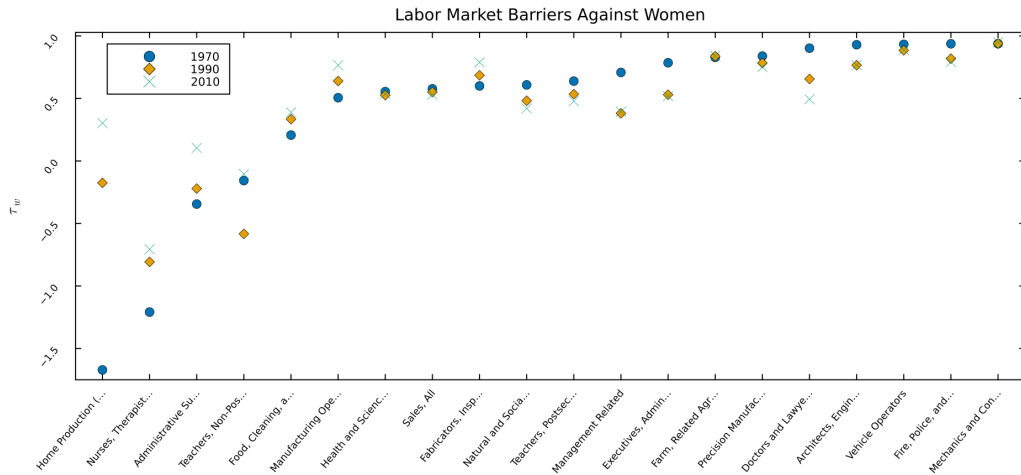
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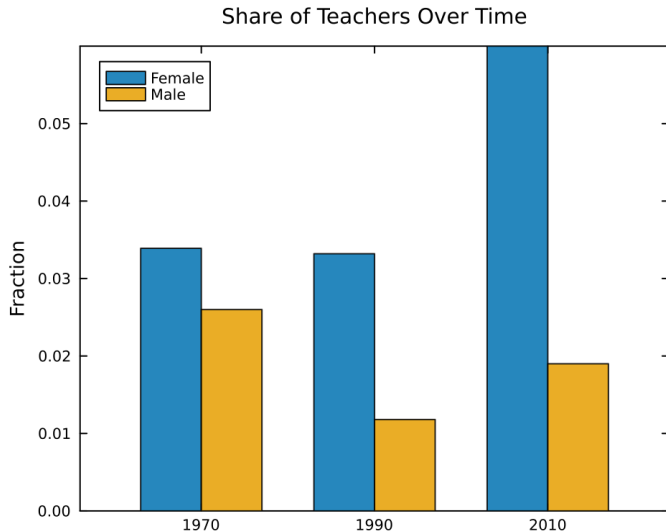


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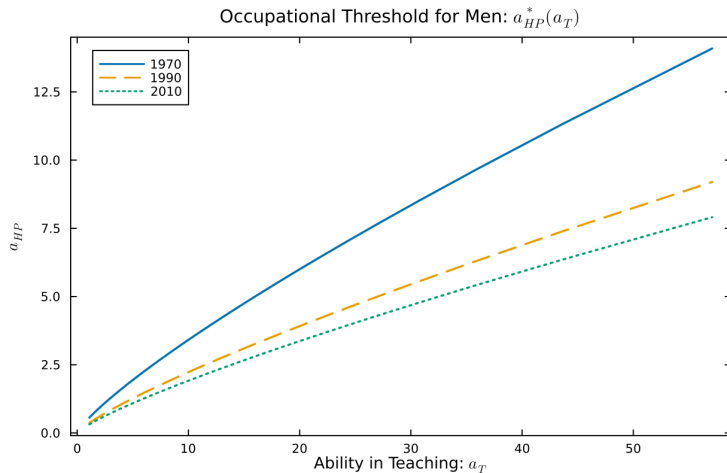


Number and Composition of Teachers



Occupational Threshold

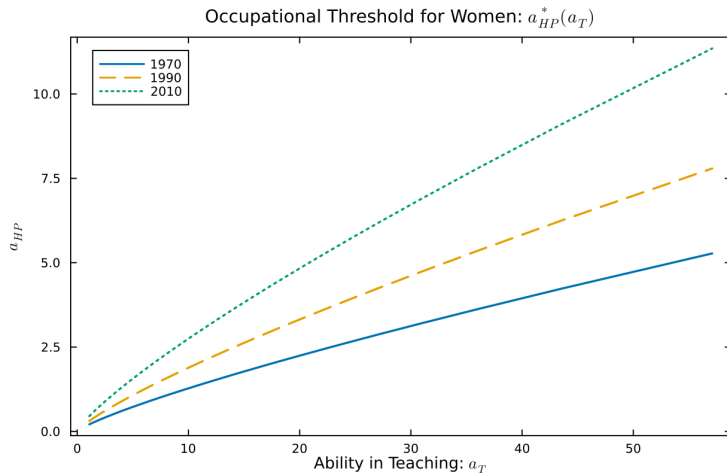
Teaching vs. Home Production, **Men**



Men with $a_{HP} < a_{HP}^*(a_T)$ become teachers.

Occupational Threshold

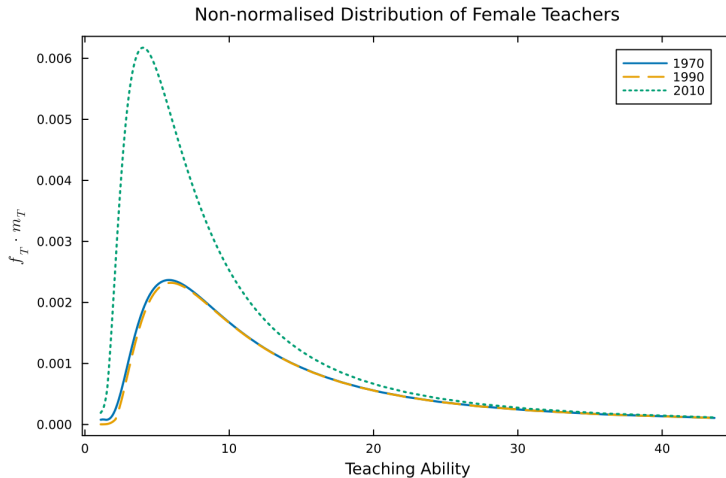
Teaching vs. Home Production, **Women**



Women with $a_{HP} < a_{HP}^*(a_T)$ become teachers.

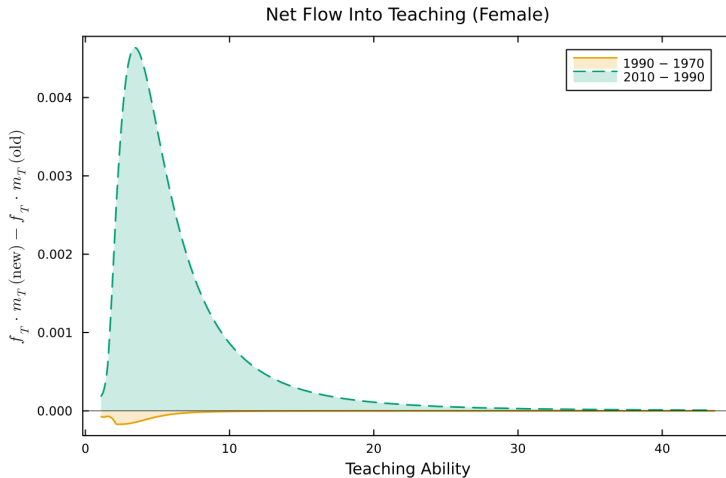
Distribution of Teaching Abilities

More women with lower abilities become teachers after 1990



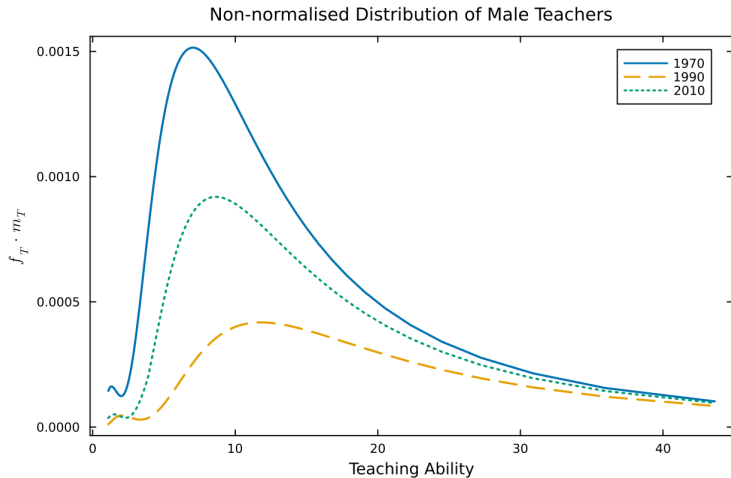
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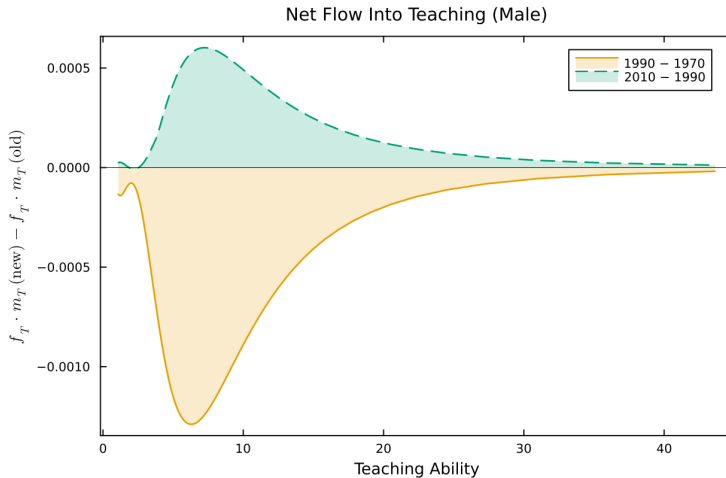
Distribution of Teaching Abilities

Men exit teaching 1970-90, partially return 1990-2010



Distribution of Teaching Abilities

Men exit teaching 1970-90, partially return 1990-2010



Human Capital (relative to 1970)

	1970	1990	2010
measure of teachers (female)	1	0.98	1.77
measure of teachers (male)	1	0.45	0.73
$\tilde{H}_T$ per teacher (female)	1	1.44	1.38
$\tilde{H}_T$ per teacher (male)	1	2.10	2.29
$\tilde{H}_T$ (aggregate)	1	1.19	2.08
H_O per worker (female)	1	1.04	1.38
H_O per worker (male)	1	1.35	1.99
H_O (aggregate)	1	1.56	2.02

Summing up

- ▶ Ability distribution and gender composition of teachers changes over time:
 - More women select teaching, ability distribution becomes more right-skewed (relative to 1970 and relative to men) ▶ Density Function for Women
 - Fewer men select teaching, distribution becomes (slightly) more left-skewed (relative to 1970) ▶ Density Function for Men
- ▶ Home production becomes relatively unattractive for women (τ_{HP} rises sharply between 1970 and 2010):
 - Higher-ability women enter non-teaching occupations
 - Lower-ability women enter teaching
 - Female workers (outside teaching) have 30% more human capital in 1970, on average, but exhibit less growth over next 40 years compared to male workers

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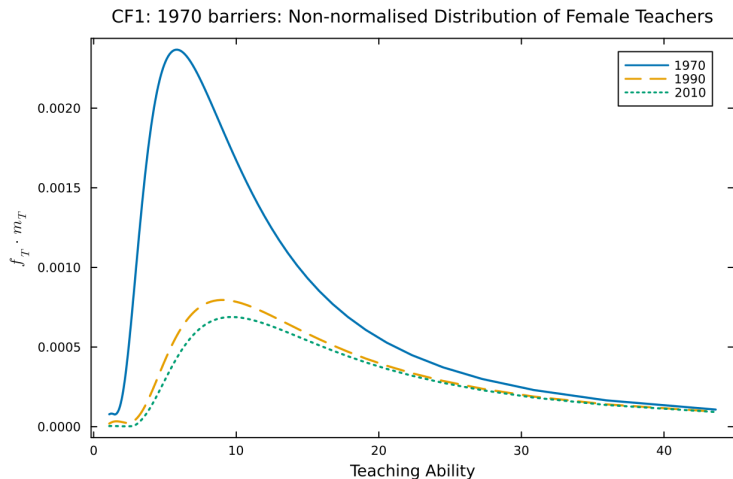
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Counterfactual Experiment #1:

Labor Market Barriers for Women Frozen in 1970

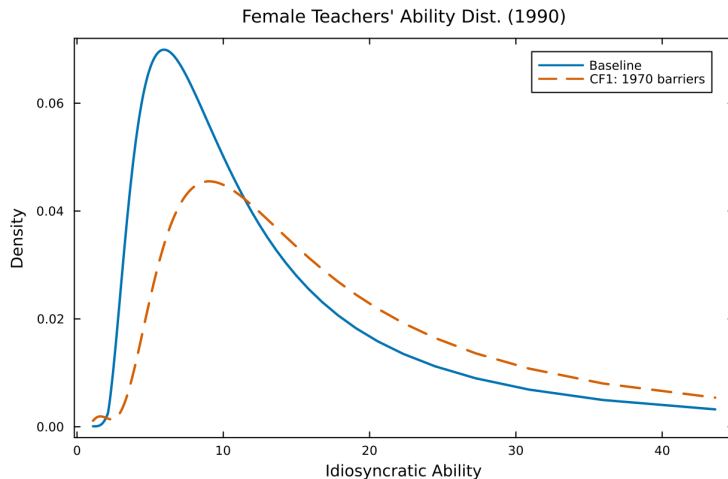
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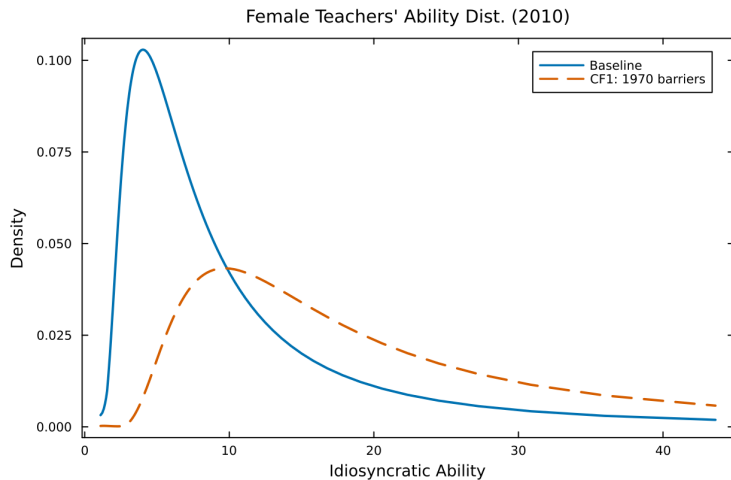
Density of Teaching Abilities vs. Benchmark



Counterfactual Experiment #1:

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Density of Teaching Abilities vs. Benchmark



Counterfactual Experiment #1:

Labor Market Barriers for Women Frozen in 1970

Human Capital: Counterfactual #1 vs. Benchmark

	1970	1990	2010
measure of teachers (female)	1	0.52	0.27
measure of teachers (male)	1	1.08	1.11
$\tilde{H}_T$ per teacher (female)	1	1.16	1.48
$\tilde{H}_T$ per teacher (male)	1	0.80	0.67
$\tilde{H}_T$ (aggregate)	1	0.71	0.53

Counterfactual Experiment #1:

Labor Market Barriers for Women Frozen in 1970

Summing Up

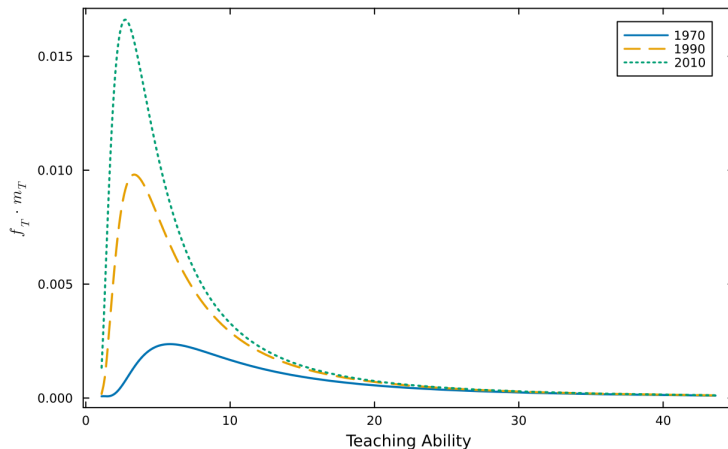
- ▶ Number of teachers and $\tilde{H}_T$ lower than in benchmark
- ▶ Ability distribution of female teachers more *right*-skewed and women's average h_T higher compared to benchmark
- ▶ Ability distribution of male teachers more *left*-skewed and men's average h_T lower compared to benchmark

Counterfactual Experiment #2:

κ and Barriers Except **Home Production** Frozen in 1970

Distribution of Teaching Abilities

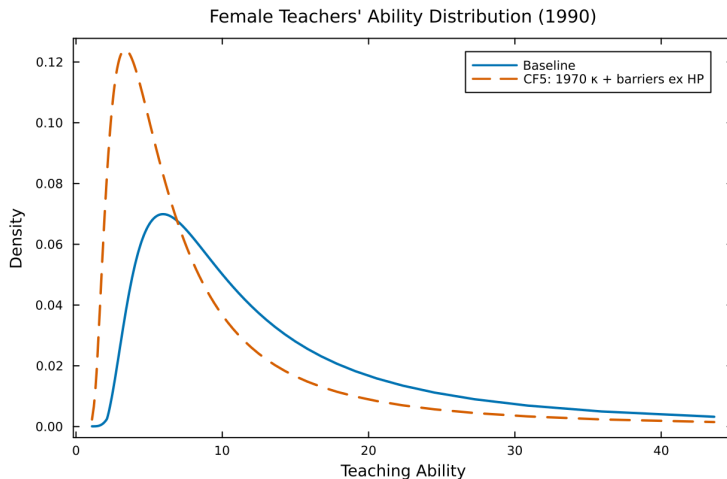
CF5: 1970 κ + barriers ex HP: Non-normalised Distribution of Female Teacher



Counterfactual Experiment #2:

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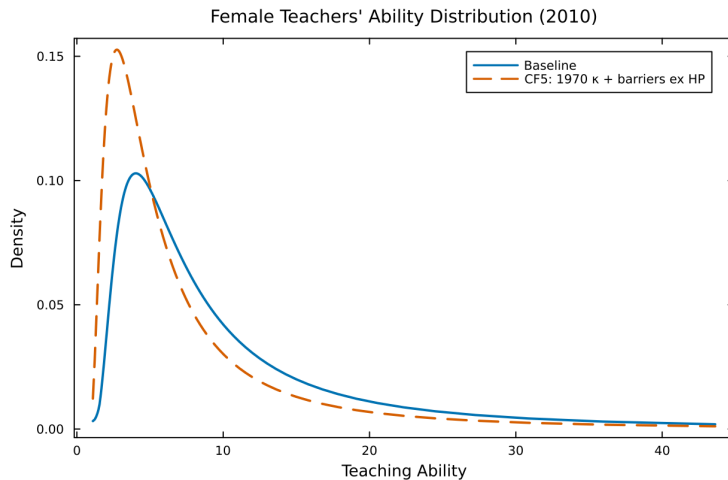
Density of Teaching Abilities vs. Benchmark



Counterfactual Experiment #2:

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Density of Teaching Abilities vs. Benchmark



Counterfactual Experiment #2:

κ and Barriers Except **Home Production** Frozen in 1970

Human Capital: Counterfactual #1 vs. Benchmark

	1970	1990	2010
measure of teachers (female)	1	2.33	1.81
measure of teachers (male)	1	2.4	2.13
$\tilde{H}_T$ per teacher (female)	1	1.08	1.06
$\tilde{H}_T$ per teacher (male)	1	1.16	1.02
$\tilde{H}_T$ (aggregate)	1	2.61	2.02

Counterfactual Experiment #2:

κ and Barriers Except Home Production Frozen in 1970

Summing Up

- ▶ Number of teachers (male and female) and $\tilde{H}_T$ higher than in benchmark
- ▶ Average h_T of teachers similar to benchmark
- ▶ Ability distribution of all teachers more *right*-skewed compared to benchmark

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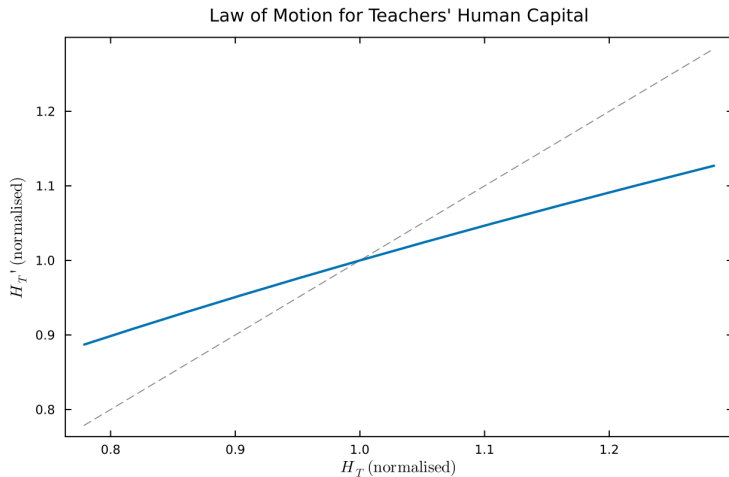
- ▶ Develop a novel theory of occupational choice and human capital formation:
 - non-linear wages
 - intergenerational dynamics of human capital accumulation
- ▶ Quantitative model characterizes compositional changes in abilities at occupation level

Conclusion

Ongoing and Future Work

- ▶ More granular decomposition of labor productivity growth into selection and human capital investment effects
- ▶ Disentangle role of $\tilde{H}_T$ and individual teachers' h_T more sharply
- ▶ Quantify extent of occupational mismatch using aptitude test scores in Project TALENT, NLSY79, NLSY97, and occupational skill requirements in O*NET
- ▶ Explore effect of competition for teaching talent across multiple locations / school districts (differentiated by amenities)

Aggregate Law of Motion



◀ Back

Optimal Time Investment

$$s_{T,g} = \frac{\mu\phi}{\mu\phi + \frac{1}{\gamma} - \eta}$$
$$s_{O,g} = \frac{\mu\phi}{\mu\phi + 1 - \eta}$$

◀ Back

Optimal Goods Investment

$$e_{T,g} = \left((\kappa \cdot \gamma \cdot \eta \cdot \tau_{T,g})^{\frac{1}{\gamma}} \cdot a_T^\alpha \cdot s_{T,g}^\phi \cdot \left(\frac{2\tilde{H}_T}{M} \right)^\sigma \right)^{\frac{1}{\frac{1}{\gamma} - \eta}}$$
$$e_{O,g} = \left(A'_O \cdot \eta \cdot \tau_{O,g} \cdot a_O^\alpha \cdot s_{O,g}^\phi \cdot \left(\frac{2\tilde{H}_T}{M} \right)^\sigma \right)^{\frac{1}{1 - \eta}}$$

where

$$\tau_{i,g} = \frac{(1 - t)(1 - \tau_{i,g}^\omega)}{1 + \tau_{i,g}^e}$$

◀ Back

Aggregate Law of Motion

$$\tilde{H}'_T = \left[(\kappa \cdot \gamma \cdot \eta)^{\frac{\eta}{1-\eta\gamma}} \cdot \left(\frac{2\tilde{H}_T}{M} \right)^{\frac{\sigma}{1-\eta\gamma}} \times \sum_{g=1}^G \tau_{T,g}^{\frac{\eta}{1-\eta\gamma}} \cdot \int_0^\infty s_{T,g}^{\frac{\phi}{1-\eta\gamma}} \cdot a^{\frac{\alpha}{1-\eta\gamma}} f_{T,g}(a) da \right]^{\frac{\beta}{\sigma}}$$

where

$$\tau_{i,g} = \frac{(1-t)(1-\tau_{i,g}^\omega)}{1+\tau_{i,g}^e}$$

◀ Back

Transition Dynamics

- ▶ Occupational choice boundary depends on $\tilde{H}_T$:

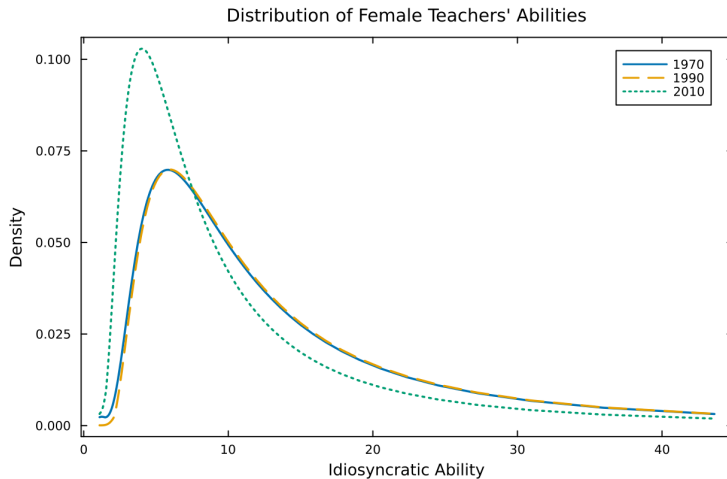
$$\frac{a_{T,g}^*(a_O)^{\frac{\alpha}{\gamma-\eta}}}{a_O^{\frac{\alpha}{1-\eta}}} \cdot \frac{s_{T,g}^{\frac{\phi}{\frac{1}{\gamma}-\eta}}}{s_{O,g}^{\frac{\phi}{1-\eta}}} \cdot \frac{\tau_{T,g}^{\frac{1}{1-\eta\gamma}}}{\tau_{O,g}^{\frac{1}{1-\eta}}} \cdot \frac{1 + \tau_{T,g}^e}{1 + \tau_{O,g}^e} \cdot \left(\frac{1 - s_{T,g}}{1 - s_{O,g}} \right)^{\frac{1}{\mu}} \\ \times \frac{(\kappa \cdot \gamma)^{\frac{1}{1-\eta\gamma}}}{A'_O{}^{\frac{1}{1-\eta}}} \cdot \frac{\frac{1}{\gamma} - \eta}{1 - \eta} \cdot \eta^{\frac{\eta(\gamma-1)}{(1-\eta)(1-\eta\gamma)}} \cdot \left(\frac{2\tilde{H}_T}{M} \right)^{\frac{\sigma(\gamma-1)}{(1-\eta)(1-\eta\gamma)}} = 1$$

where

$$\tau_{i,g} = \frac{(1-t)(1-\tau_{i,g}^\omega)}{1+\tau_{i,g}^e}$$

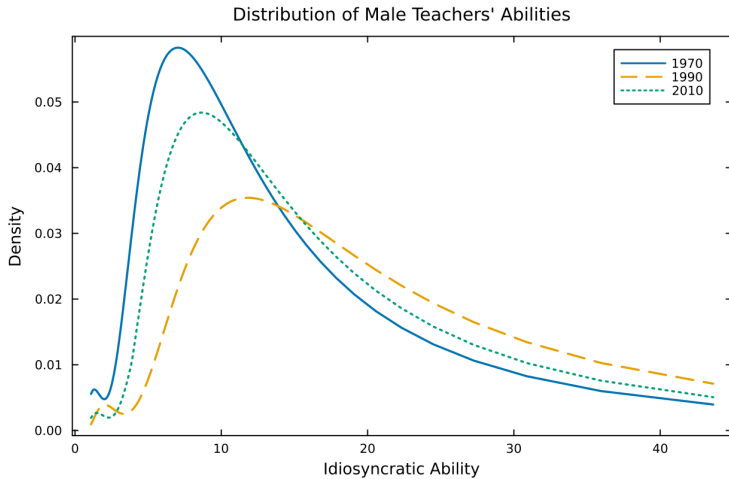
- Relationship between $\tilde{H}_T$ and $a_{T,q}^*(a_O)$ “drives” transition dynamics

Density Function for Ability of Female Teachers



◀ Back

Density Function for Ability of Male Teachers



◀ Back

List of Occupations

1. Executives, Administrative, and Managerial
2. Management Related
3. Architects, Engineers, Math, and Computer Science
4. Natural and Social Scientists, Recreation, Religious, Arts, Athletes
5. Doctors and Lawyers
6. Nurses, Therapists, and Other Health Service
7. Teachers, Postsecondary
8. Teachers, Non-Postsecondary and Librarians
9. Health and Science Technicians
10. Sales, All
11. Administrative Support, Clerks, Record
12. Fire, Police, and Guards
13. Food, Cleaning, and Personal Services and Private Household
14. Farm, Related Agriculture, Logging, and Extraction
15. Mechanics and Construction
16. Precision Manufacturing
17. Manufacturing Operators
18. Fabricators, Inspectors, and Material Handlers
19. Vehicle Operators
20. Home Production

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